Challenge Brief

Template for challenge partners · Since AI Hackathon 2026 · 6–8 Nov 2026, Turku

This template helps you define a clear, buildable challenge for the hackathon. Fill in each section below — the more concrete the problem and the evaluation criteria, the better the prototypes you receive. You can return it to us as is, and we are happy to help shape it into a ready event challenge.

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Challenge partner (company)

Transport and Communications Agency Traficom / Spectrum and Space -department

01 Challenge Title

A short, memorable name for the challenge. Example: “Automated support ticket routing.”

Saving Earth from Space Interferers Using Radio Frequencies

Develop an analysis tool to protect national radio frequency use from planned space and terrestrial usage to prevent potential harmful interferences.

02 Problem Statement

What business or technical problem should be solved? Describe the context, why it matters, and the main pain points.

Using radio frequencies ('spectrum') requires preliminary planning and obeying International Telecommunications Union (ITU) rules ('International Radio Regulations'). The usage differs in different countries and regions. When using the spectrum in space (i.e. satellites) the challenge becomes global: Australian terrestrial usage does not interfere Finnish usage, but Australian satellite orbiting the Earth may cause interference to Finnish terrestrial usage in the same or adjacent frequency range.

Due to this, ITU's Radio Bureau ('BR') International Frequency Information Circular ('IFIC') publications. The publications are available for space and terrestrial use separately, containing bi-monthly update on both pdf and database format.

To analyse these, specialists need to identify items that may be relevant to Finland, compare them with other frequency information (especially the table of frequencies in Finland, but also the list of frequencies of special interest). When possible challenges are identified, the specialist assesses possible technical concerns and prepare follow-up correspondence to other administration planning the usage (for solving the situation (technically) before it happens, or to preserve rights to claim protection in case of realised harmful interference case).

The challenge is to build an AI-assisted workflow that can reduce the manual effort, enhance analysis coverage and increase the quality of analysis, while keeping the analysis transparent and technically reviewable. A first demonstrator may focus on the space-services side and should be designed so that the same architecture could later be extended to terrestrial frequency-planning workflows.

A possible end-to-end workflow includes:

- ingesting relevant national information of frequency usage (regulations, additional frequency tables provided);
- detecting, retrieving and ingesting [new] ITU BR IFIC publications, especially API/A, CR/C and AP30 (including AP30A and AP30B);
- possible using ITU SpaceCom tool (or direct database logic) to identify interesting satellite systems;
- possible using ITU Gibc tool (or direct calculation logic) to calculate pfd values of the satellite systems;
- summarising the relevant content and frequency ranges affecting Finland or Finnish usage;
- cross-comparing the findings with information on national regulations ('M4', 'M15') and additional frequency tables;
- identifying items that may concern Finland based on both previous public material and possible synthetic data of frequency, pfd limits, and geographical locations;
- detecting and explaining potential concerns, including technical calculations on f. ex. power-flux density (pfd), where appropriate;
- identifying the relevant counterparty (administration) and drafting relevant correspondence template for expert review where possible interference concern is described.

Side quest: Terrestrial land-mobile notices

Demonstrate the architecture on terrestrial BR IFIC data limited to terrestrial land-mobile notices. The terrestrial side quest should show that the same workflow, with additions, modifications, or simplifications defined below, can support a different context without turning the submission into a second full challenge.

- Use consecutive terrestrial BR IFIC issues and identify what is new, modified or withdrawn;
- filter foreign assignments that may be relevant to Finland using notice data;
- determine the applicable ITU, CEPT/ECC and national regulatory framework;
- based on applicable regulatory framework, identify the applicable coordination or protection criterion such as ECC T/R 25-08;
- using the coordination or protection criteria, use deterministic method to calculate field strengths caused by the notified station. Use ITU-R P.1546 propagation model; and
- produce expert reviewable findings showing the source notice, calculated field strengths, model parameters, assumptions and source citations.

Additionally, surface a small number of trends or anomalies in neighboring countries administrations’ notices, such as new activity by band, geography, station class or bandwidth.

03 Guiding Question

One question that focuses the challenge. Example: “How might we use AI to predict equipment failures before they happen?”

How might we build an auditable AI-assisted workflow that automatically monitors ITU BR IFIC information, identifies cases relevant to Finland, supports technical (frequency and radio frequency power level) analysis, and prepares expert-reviewable draft correspondence?

GSO and non-GSO satellite power flux densities are in special interest to recognize possible interfering usage, as well understanding trends and planned usage of new satellites systems which may interfere Finnish satellites already in use. Identifying these issues from huge mass of IFIC API/A, CR/C, and AP30 (AP30A and AP30B) information is essential, and there AI solution might be helpful.

To successfully complete this task, you need to take into account available information on the Finnish radio frequency usage (i.e. Regulation 'M4' and additional frequency tables of special interest provided on the challenge platform) which are used on the comparisons and possible interference recognition. Additional data may be used, for example harmonised use of radio frequencies in Europe or other public radio frequency data usage sources, to raise further notes and concerns on the IFIC information.

04 Recommended AI Approaches *Optional*

Any relevant technologies, models or methods — for example NLP for text analysis, classification, knowledge graphs, agents, or RAG.

Possible approaches may include document and structured-data ingestion directly from public source, information extraction, retrieval-augmented generation (RAG), rule-based or deterministic calculation components, agentic workflows, anomaly detection, ranking and prioritisation, and auditable citation or provenance mechanisms. Teams are not required to use any specific AI architecture; technical correctness, traceability and expert reviewability are more important than the choice of model.

05 Dataset or API Provided *If any*

Describe any data or resources you provide: format, a link if ready, and sample size. If none, state what builders should expect.

> At Since AI, challenges can be built with synthetic, anonymized or public data — no personal data is required.

NOTE: Datasets beginning with '**' are mandatory and their usage is needed to successfully complete this challenge. All other data may be used (listed here or from other sources).

** ITU Space IFIC: https://www.itu.int/en/ITU-R/space/brific/Pages/default.aspx
- Downloadable Space IFIC: https://www.itu.int/sns/wic/demowic.html
- All existing IFICs and forthcoming IFICs shall be able to be evaluated
- Space IFIC no. 3074 needs to be analysed and possible concerns shall be raised from it (3074 is used to evaluate accuracy, and results from it shall be separately submitted)

** ITU Terrestrial IFIC: https://www.itu.int/en/ITU-R/terrestrial/brific/Pages/default.aspx
- Downloadable Terrestrial IFIC extract: https://www.itu.int/en/ITU-R/terrestrial/brific/Pages/default.aspx
- All existing IFICs and forthcoming IFICs shall be able to be evaluated
- Terrestrial IFIC no. 3076 needs to be analysed and possible concerns shall be raised from it (3076 is used to evaluate accuracy, and results from it shall be separately submitted)

** Regulation 4 ('M4', Finnish Radio Frequency Regulation, including table of frequencies in Finland): https://www.finlex.fi/fi/viranomaiset/maarayskokoelmat/traficom-viestinta/2026/3
- Download radio frequency table from API: https://opendata.traficom.fi/swagger/ui/index#/Taajuusjakotaulukko - https://opendata.traficom.fi/api/v13/Taajuusjakotaulukko

** Additional dataset (XLS-file) containing synthetic dataset of a table of frequencies of special interest, including station number, frequency range (bandwidth and center frequency), terrestrial coordinates, technical information (antenna height, antenna direction, antenna directivity, etc.). This dataset is available on SinceAI Platform.

ITU tools: https://www.itu.int/en/ITU-R/software/Pages/space-network-software.aspx
- These tools can be used to handle and analyse IFIC datasets

Regulation 15 ('M15'), Collective Frequencies for Licence-exempt Radio Transmitters and Their Use (in Finland)): https://www.finlex.fi/fi/viranomaiset/maarayskokoelmat/traficom-viestinta/2025/51209 (this information not necessarily as relevant as previous one)

CEPT implementation status and material of frequency harmonization in Europe: https://docdb.cept.org/implementation/overview
- This data covers harmonised radio frequency use in Europe, including in Finland, and their technical harmonisation parameters

ITU Radio Regulations: https://www.itu.int/pub/r-reg-rr
- This is the international agreement of the radio frequency usage, and it contains the limits how they may be used, especially when radio frequency transmitter causes power levels cross-border to another country, or satellite uses frequencies above another country

06 Evaluation Criteria

How should solutions be judged? What matters most — for example accuracy, scalability, usability, or business impact?

A. Technical analysis and reproducibility — 25%
- technically credible calculations and reasoning
- deterministic and reproducible components where appropriate
- assumptions and model parameters are visible

B. Extensibility and terrestrial side quest — 25%
- demonstrates reuse of the core architecture
- shows credible extension from space services to terrestrial land-mobile notices

C. Workflow usefulness, innovation and output quality — 20%

– prioritisation of findings is useful to an expert
– outputs are understandable, reviewable, and well documentable
– innovation adds useful new ideas while remaining realistically implementable

D. Case recognition accuracy and relevance — 15%
– correctly identifies cases relevant to Finland and Finnish terrestrial use
– minimises material false positives and false negatives

E. Auditability and evidence quality — 15%
– clear source provenance
– citations and references to source notices, regulations or standards
– an expert can understand why a conclusion was reached

TOTAL: 100%

07 Submission Format

What deliverables do you expect? For example a demo video, pitch slides, a code repository, or documentation.

- Demo of the solution (preferably some functionality)
- Ability to process IFIC files generating relevant output (using the Finnish usage information)
- Processed Space IFIC 3074 and Terrestrial IFIC 3076 for accuracy evaluation purposes (separate file(s) to be provided)
- Information on the input/output leading to the solution
- Code repository of the solution
- Other clarifying documentation is suggested, as it helps to evaluate the solution

08 Point of Contact

Name, email and preferred contact method for questions during the hackathon.

Discord is the preferred contact method.

Contact point of the Q&A stand:
Traficom SinceAI 2026, Traficom.sinceai.2026@gmail.com, Discord: traficomsinceai2026

Contact point of the specialists on the challenge:
Pekka Pussinen, pekka.pussinen@traficom.fi, tel. +358 29 539 0454, Discord: pekkatraficom_83563
Otto Julkunen, otto.julkunen@traficom.fi, tel. +358 295 346 438, Discord: ottoj0436_45333

Available on Q&A stand: Sat 9.00-12.00 and Sat 14.30-18.00 (or as requested and agreed)

Senior specialists on the IFICs and processes are available - if previously mentioned specialists are not able to answer questions - in following time slots (for answering questions / calls):

Space IFIC: Sat 10-11, 17-18
Terrestrial IFIC: Sat 11-12, 16-17